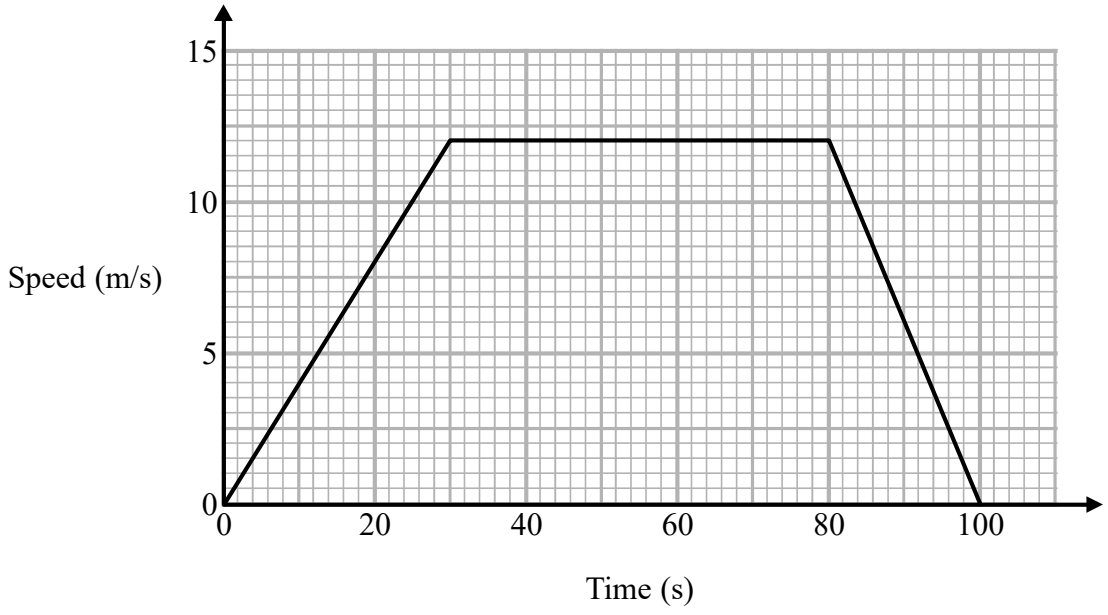
21 Here is a speed-time graph for a train journey between two stations.
The journey took 100 seconds.



(a) Calculate the time taken by the train to travel half the distance between the two stations.
You must show all your working.

Distance = area under speed–time graph.

First triangle: ½ × 30 × 12 = 180 m.

Middle rectangle: 50 × 12 = 600 m.

Last triangle: ½ × 20 × 12 = 120 m.

Total 900 m, so half is 450 m.

After 30 s, another 270 m at 12 m/s takes 22.5 s.

Half-distance time = 30 + 22.5 = 52.5 s.

4 marks total
First area
Total 900 m
Halfway through level section
52.5 s

..... seconds
(4)

(b) Compare the acceleration of the train during the first part of its journey with the acceleration of the train during the last part of its journey.

First acceleration = 12/30 = +0.4 m/s^2.

Last acceleration = (0–12)/20 = –0.6 m/s^2:

the train slows more sharply at the end.

1 mark
Opposite signs; final steeper

(1)

(Total for Question 21 is 5 marks)